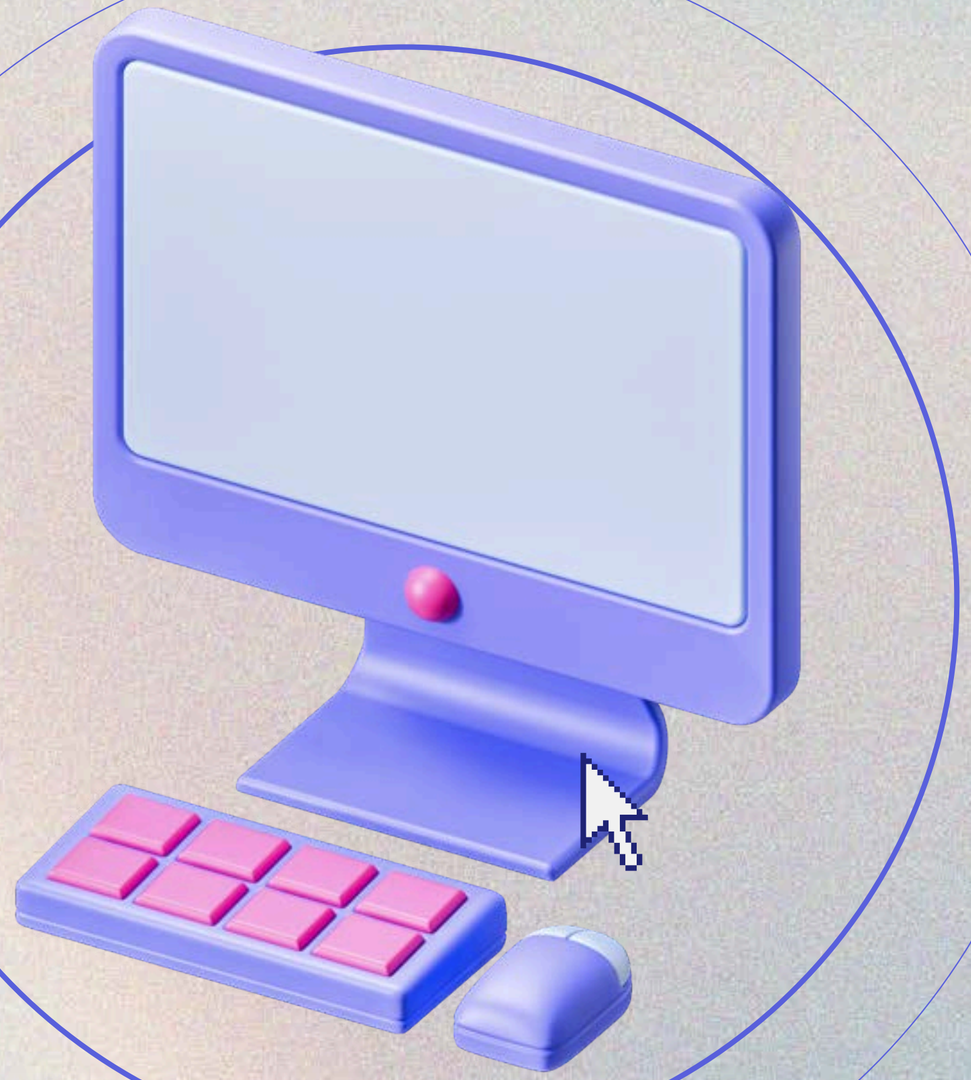
CSS3543 System Administration & Network Management

COMPUTER COMPONENT & NETWORK COMPONENT

by Zahiruddin Ibrahim



LESSON OUTCOMES

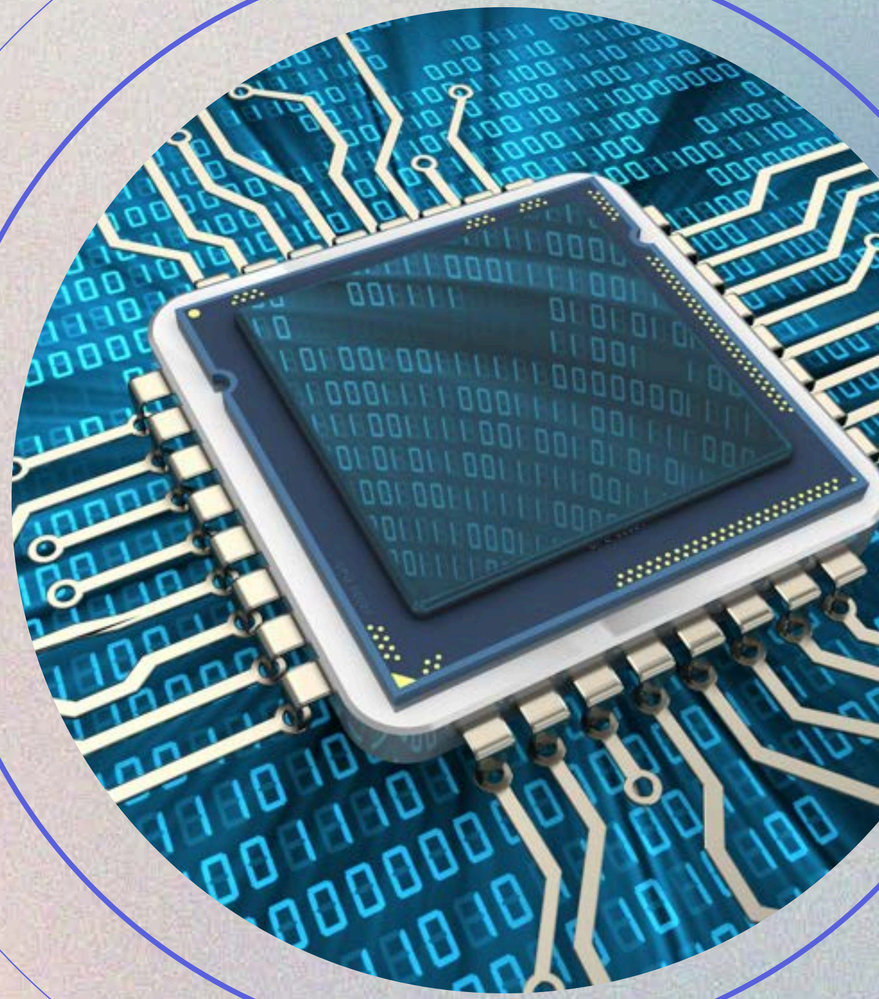
- LO1: Identify computer component including hardware, software and operating system
- LO2: Identify network component

HARDWARE IN THE SYSTEM

- Hardware refers to the physical components of a computer system.
- Two main categories:
  - Internal Hardware: CPU, RAM, Storage, Motherboard
  - External Hardware: Monitors, Keyboards, Printers, etc

CENTRAL PROCESSING UNIT (CPU)

- Executes instructions and processes data.
- Components:
  - ALU (Arithmetic Logic Unit) – Performs calculations.
  - CU (Control Unit) – Directs system operations.
- Multi-core processors vs. Single-core processors.



MEMORY HIERARCHY

- Registers (Fastest, inside CPU)
- Cache Memory (L1, L2, L3 – speeds up processing)
- RAM (Random Access Memory) (Temporary storage for active programs)
- Storage (HDD, SSD, NVMe) (Long-term data storage)

STORAGE DEVICES

- HDD vs. SSD:
  - HDD (Slower, mechanical parts)
  - SSD (Faster, flash memory, no moving parts)
- NVMe (Newer technology, much faster than SSDs) (focus more to protocol-framework)
- Cloud Storage (Google Drive, OneDrive, etc.)

OPERATING SYSTEMS

- **Operating System (OS)**
  - An OS is software that manages hardware and software resources.
  - Examples: Windows, macOS, Linux, Android.
- **Functions of OS**
  - Process Management (Scheduling tasks & running programs)
  - Memory Management (Allocating RAM to applications)
  - Device Management (Managing I/O devices)
  - File System Management (Organizing files & directories)
  - User Interface (CLI vs. GUI)

TYPES OF OPERATING SYSTEMS

- **Operating System (OS)**
  - Single-user vs. Multi-user
  - Real-time OS (RTOS) (Used in embedded systems)
  - Distributed OS (Multiple computers working together)
  - Network OS (Manages network resources)

FILE SYSTEMS

Operating System (OS)

- A filesystem organizes and manages data storage on devices.
  - Example: FAT32, NTFS, ext4, APFS.
- Filesystem Structure
  - Hierarchy of files and directories.
  - File Attributes: Name, size, creation date, permissions.
- Common Filesystems
  - Windows: NTFS, FAT32, exFAT.
  - Linux: ext4, XFS, Btrfs.
  - macOS: APFS, HFS+.

PROCESSES AND JOB CONTROL

What is a Process?

- A process is an instance of a running program.
- Processes vs. Programs:
  - A program is a set of instructions stored on disk (passive).
  - A process is an executing instance of a program (active).

PROCESSES AND JOB CONTROL

What is a Process?

- Types of Processes:
  - Foreground Processes (Interactive): Require user input (e.g., a web browser).
  - Background Processes (Daemon): Run without user interaction (e.g., system services).
- Process Characteristics:
  - Has a unique Process ID (PID).
  - Uses CPU and memory resources

JOB SCHEDULING AND PROCESS CONTROL

Job Scheduling is a process in operating systems that manages the execution of multiple processes by determining which jobs (tasks) should be executed, when they should be executed, and how system resources should be allocated to them.

TYPES OF JOB SCHEDULING

Long-Term Scheduling (Admission Scheduling)

- Decides which jobs enter the system for processing.
- Loads processes from secondary storage (e.g., hard disk) into memory.
- Balances CPU-bound vs. I/O-bound processes to optimize performance.

TYPES OF JOB SCHEDULING

Medium-Term Scheduling (Swapping)

- Temporarily removes processes from memory to manage RAM usage.
- Moves inactive processes to disk (swapping) to free memory for active processes.

TYPES OF JOB SCHEDULING

Short-Term Scheduling (CPU Scheduling)

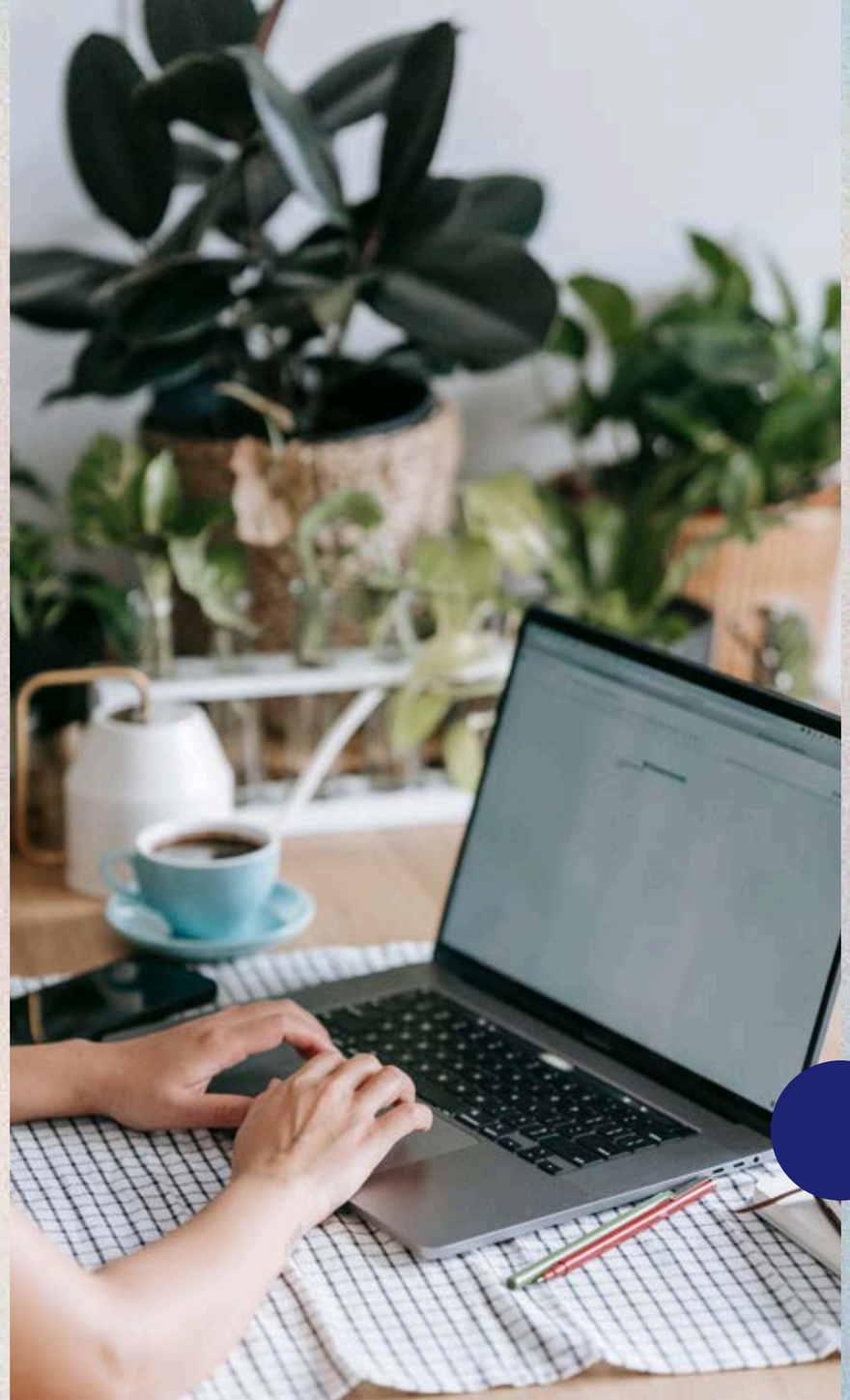
- Decides which process gets CPU time next.
- Selects processes from the ready queue and assigns CPU time.
- Uses scheduling algorithms like:
  - First-Come, First-Serve (FCFS) – Executes in order of arrival.
  - Shortest Job Next (SJN/SJF) – Runs the shortest task first.
  - Round Robin (RR) – Gives each process a time slice
  - Priority Scheduling – Runs higher-priority tasks first.

IMPORTANCE OF JOB SCHEDULING

- Ensures fair CPU usage among multiple processes.
- Optimizes system performance by balancing tasks.
- Prevents system overload by managing resource allocation.
- Improves responsiveness in multitasking systems.

NETWORK COMPONENTS

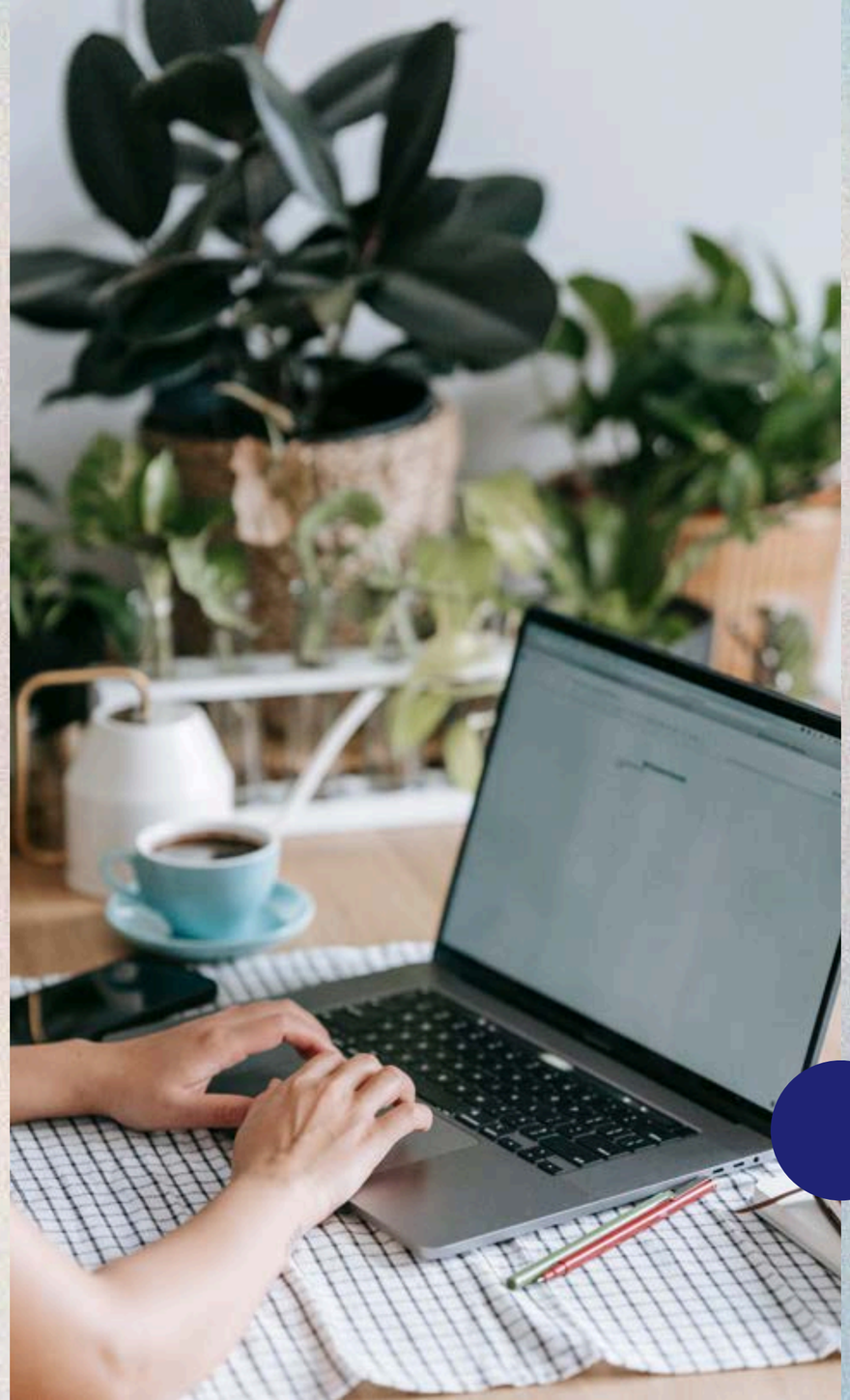
- Networking devices enable communication between computers and other devices.
- Each device has a specific role in managing data flow within a network.
- Common devices include Routers, Switches, Hubs, Modems, and Firewalls.



NETWORK COMPONENTS

Types of Network Connections:

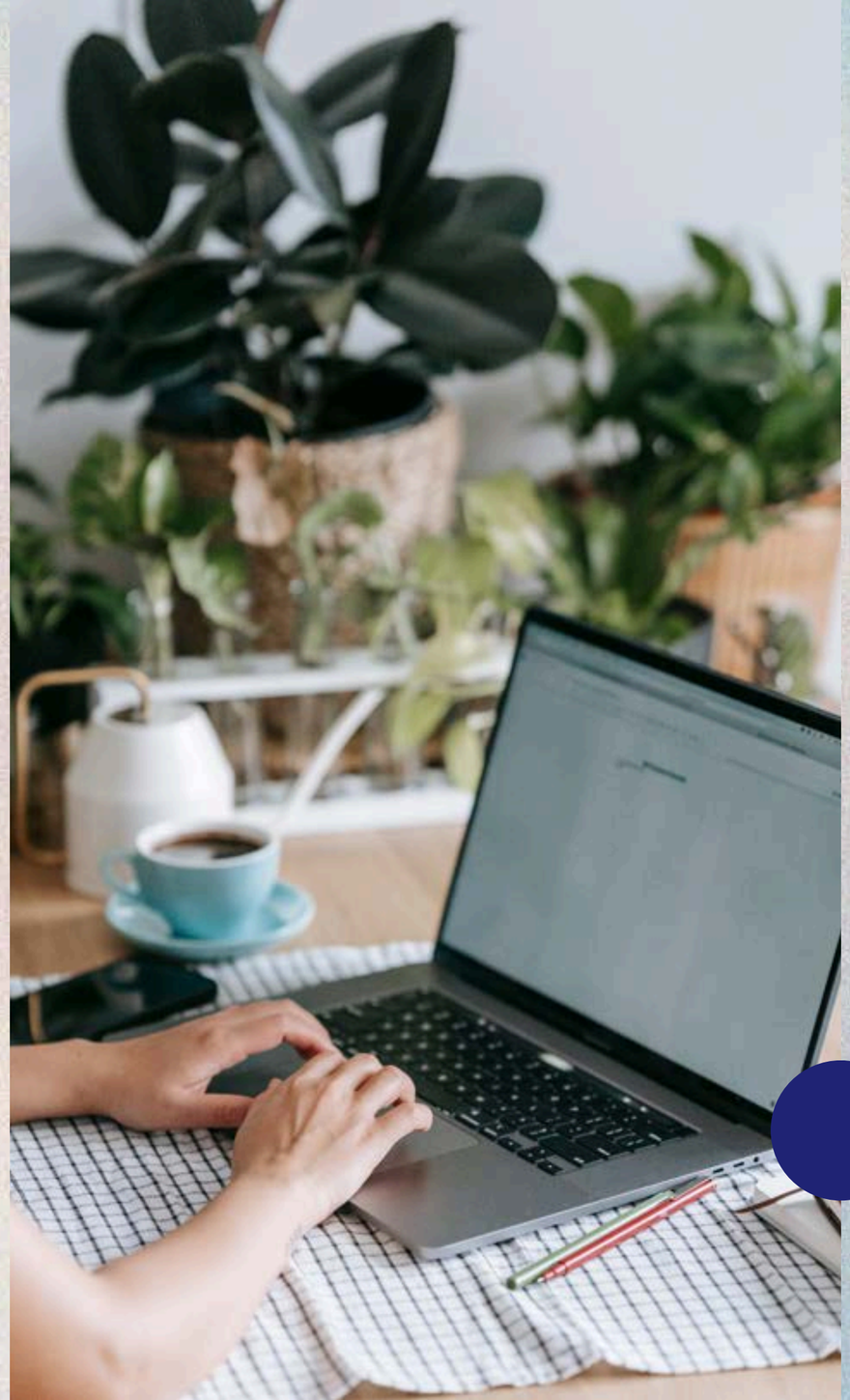
- Wired (Ethernet, Fiber Optic)
- Wireless (Wi-Fi, Bluetooth, 5G)



NETWORK COMPONENTS

Important component:

- Router: Directs traffic between networks, connects to the internet.
- Switch: Connects multiple devices in a local network.
- Hub: Broadcasts data to all devices (less efficient than a switch).
- Modem: Converts digital data for internet transmission.
- Firewall: Protects networks from cyber threats.

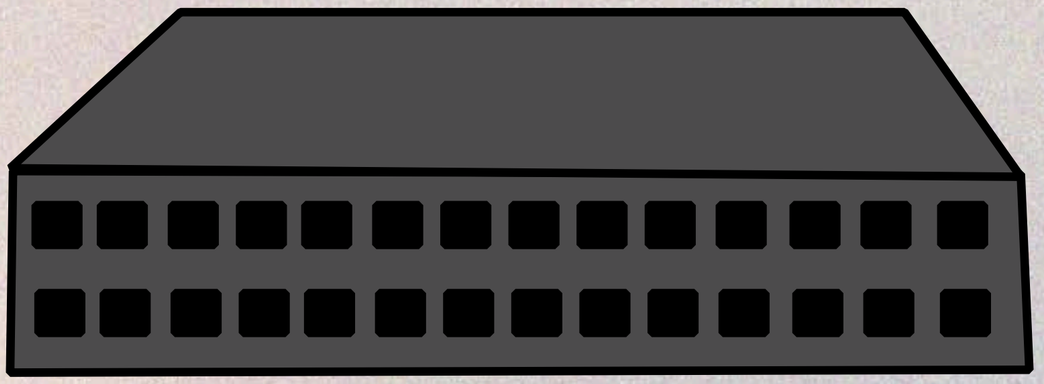


SWITCH

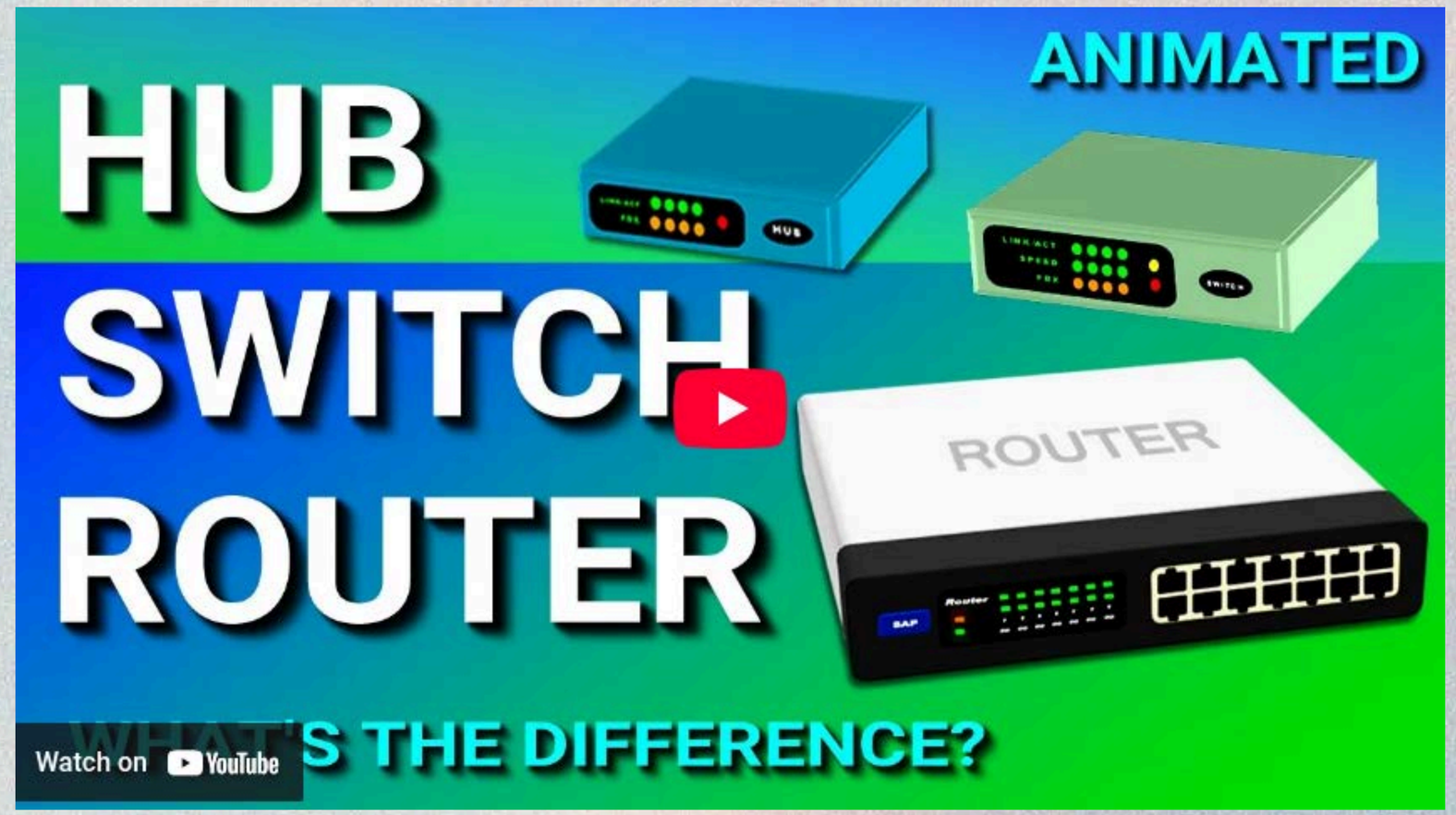
- Connects devices within a local network (LAN).
- Uses MAC addresses to forward data only to the intended device.
- More efficient than hubs (reduces network congestion).
- Example: Used in offices, data centers, and enterprise networks

HUB

- A basic networking device that connects multiple devices.
- Broadcasts data to all connected devices, causing network congestion.
- Types: Passive Hub and Active Hub.
- Example: Used in small networks but largely replaced by switches

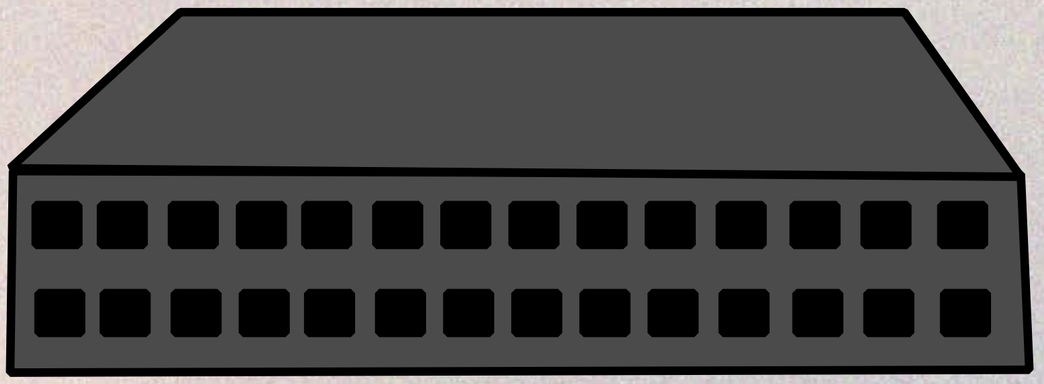


- Scan QR and watch the video provided



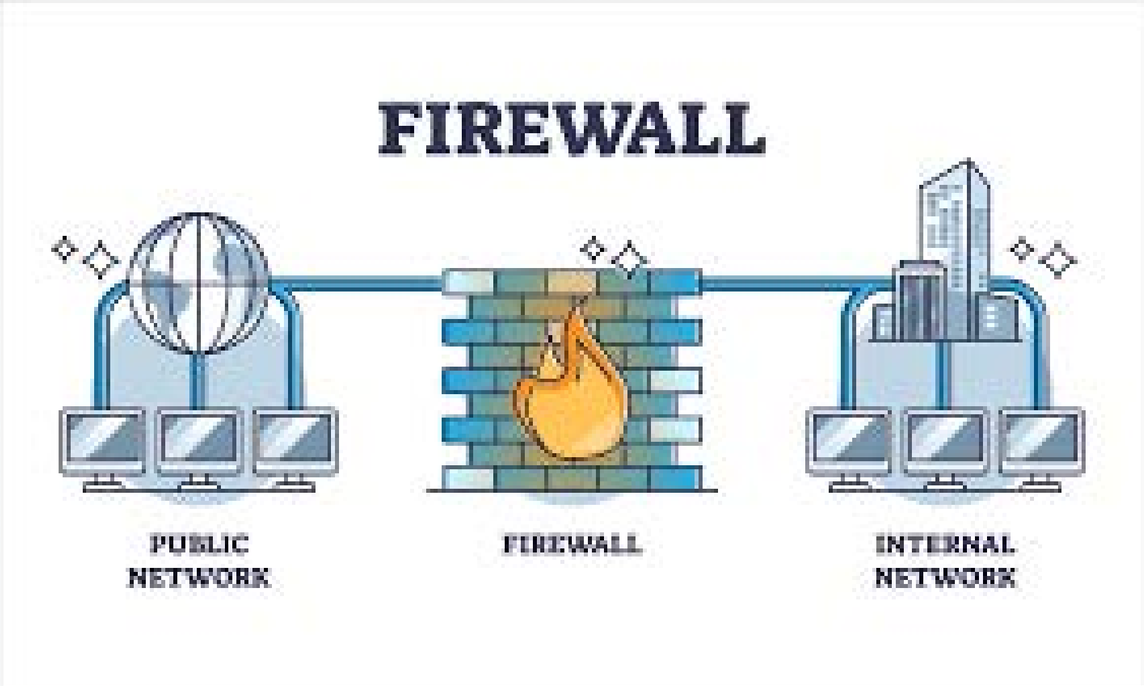
MODEM

- Converts digital signals (from a computer) into analog signals (for transmission over telephone lines) and vice versa.
- Enables internet access by communicating with an Internet Service Provider (ISP).
- Types: DSL Modem, Cable Modem, Fiber Modem.
- Example: Used in homes and offices for broadband internet access.

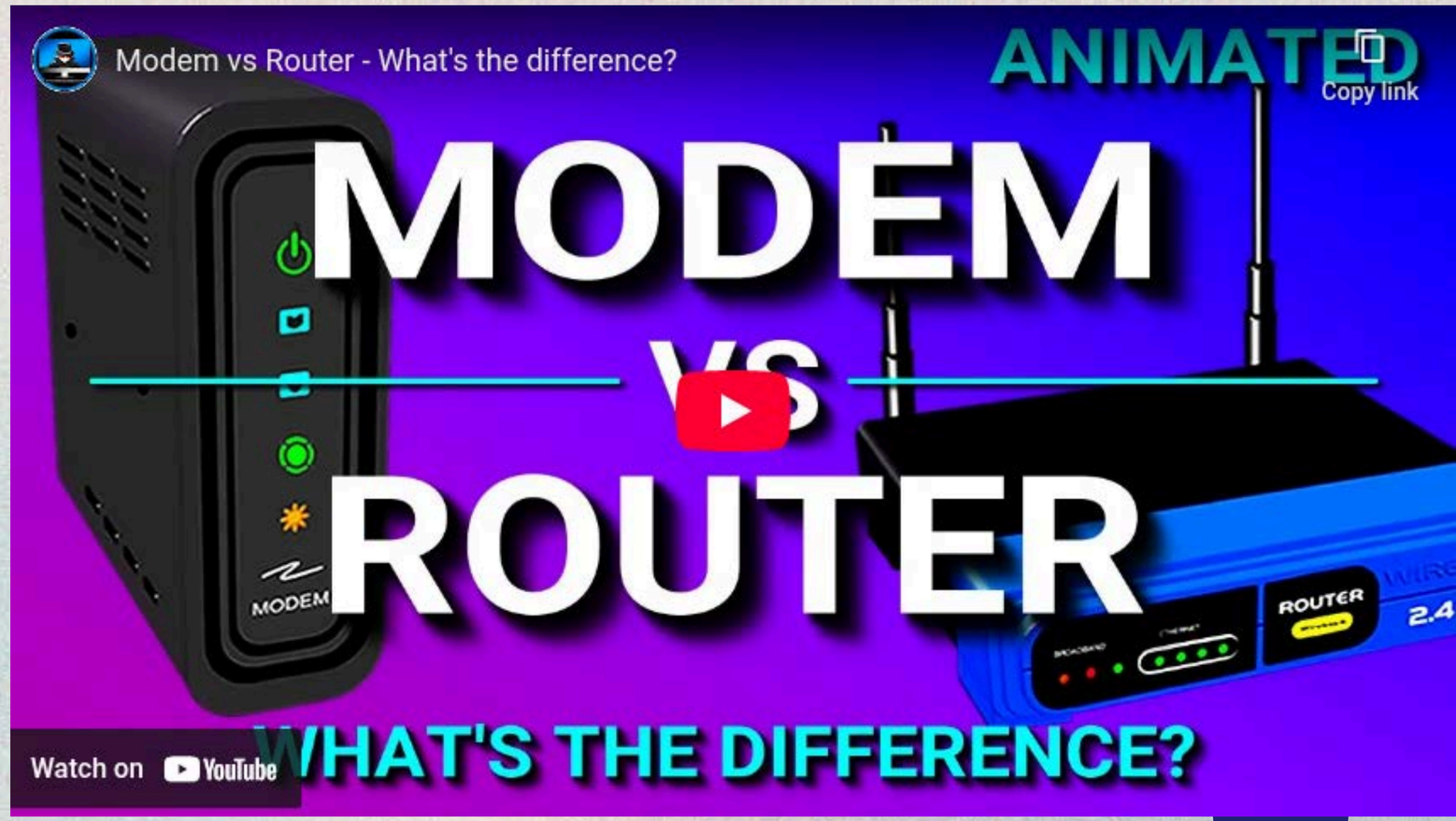


FIREWALL

- Security device that monitors and controls incoming and outgoing network traffic.
- Blocks unauthorized access while allowing legitimate communication.
- Types: Hardware Firewall, Software Firewall.
- Example: Used in businesses and personal devices to protect against cyber threats.



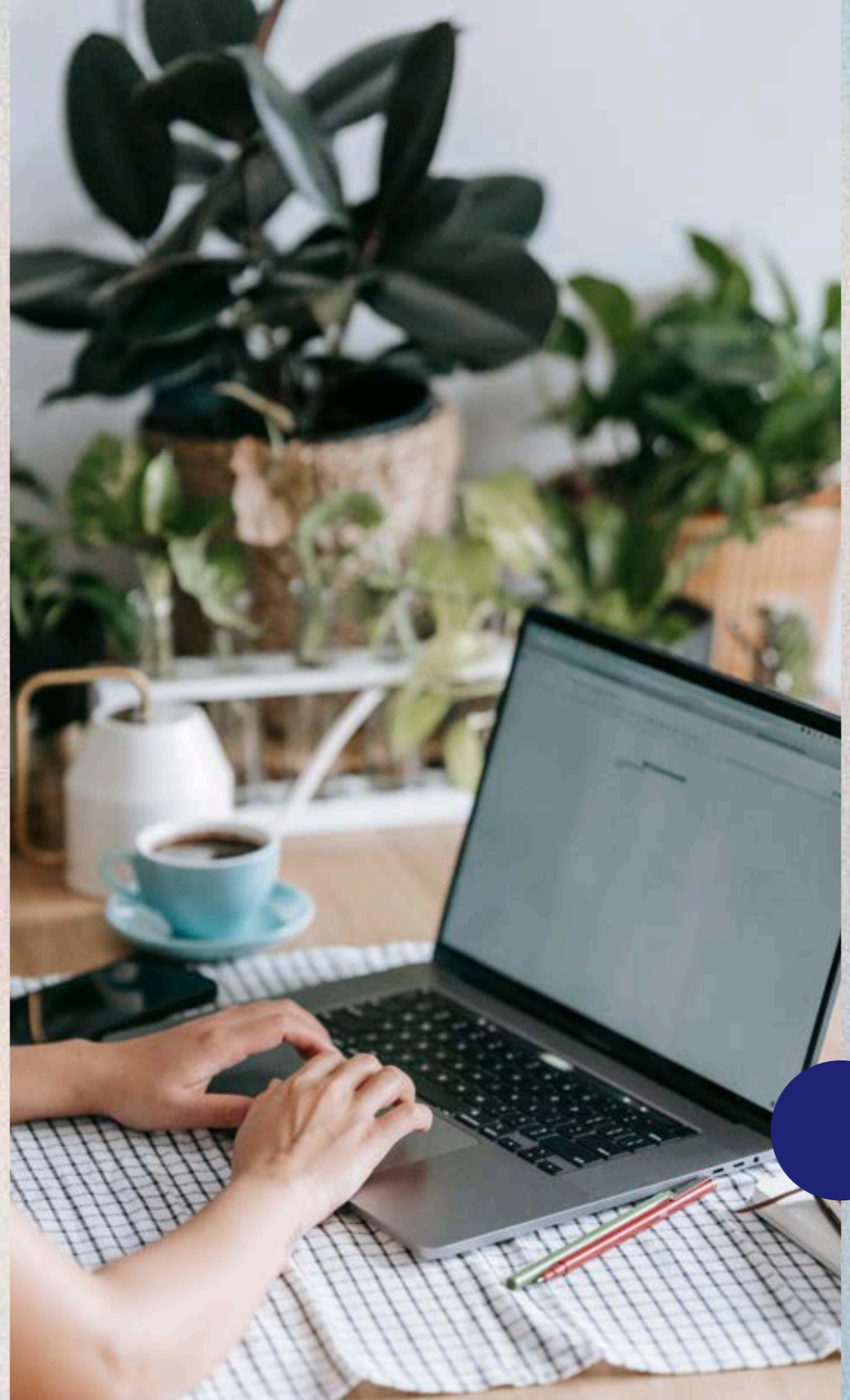
FIREWALL


Modem vs Router - What's the difference?
ANIMATED
Copy link
MODEM
VS
ROUTER
MODEM
ROUTER
WHAT'S THE DIFFERENCE?
Watch on YouTube

CLASS ACTIVITIES

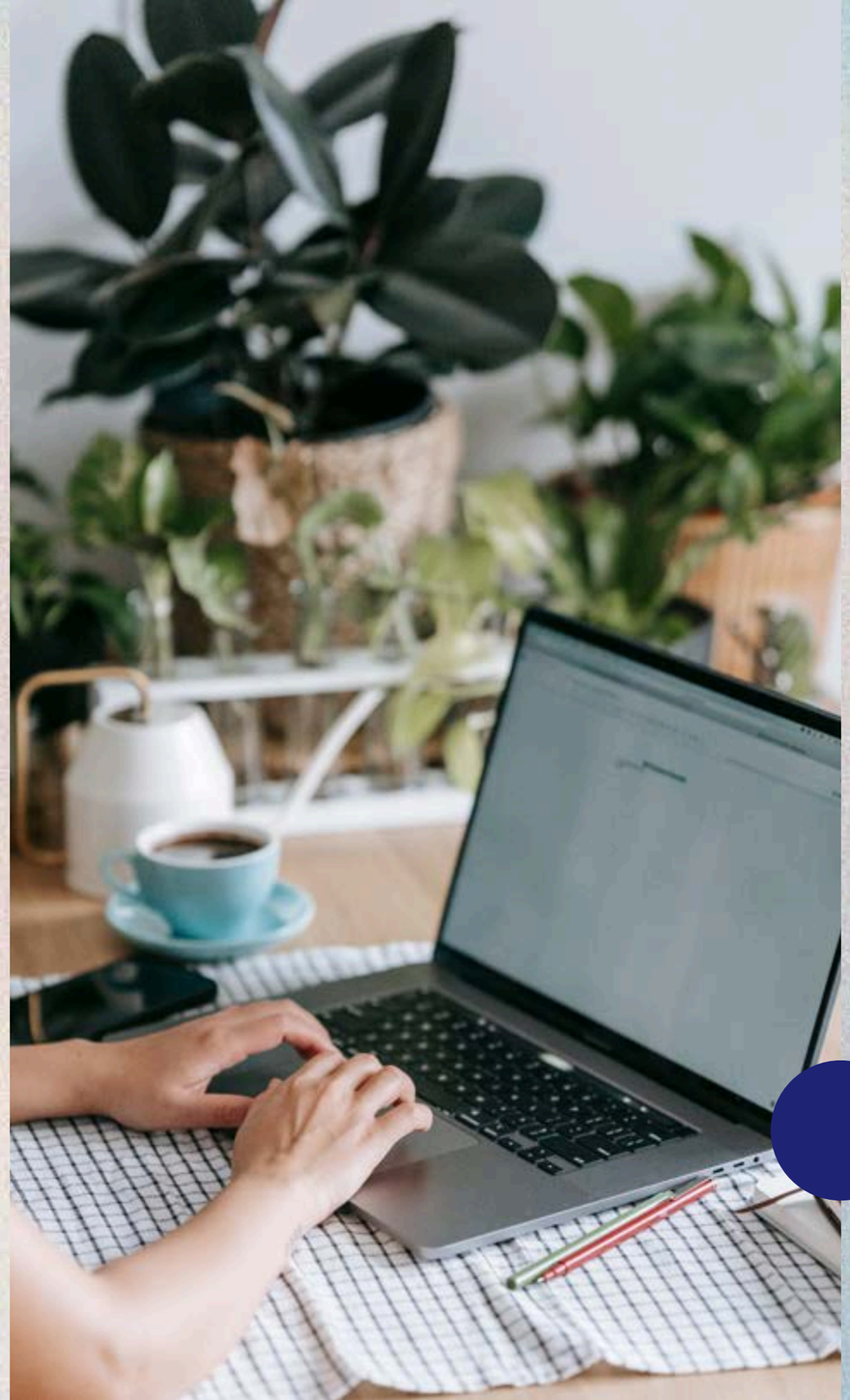
- Create a PPT slide explaining the different topologies and their advantages and disadvantages.
  - Star Topology
  - Mesh Topology
  - Bus Topology
  - Ring Topology

P.S. If I see that you are not doing the task, you will present



NETWORK TOPOLOGIES

- Star Topology
- Mesh Topology
- Bus Topology
- Ring Topology



WHAT IS AN IP ADDRESS?

- A unique numerical identifier assigned to each device in a network.
- Used for:
  - Identifying devices.
  - Routing data packets between networks.
  - Enabling internet communication.
- IP Addressing Types:
  - Static IP: Manually assigned, does not change.
  - Dynamic IP: Assigned by DHCP, changes periodically.

THANK YOU!